EE568 Project-1: Torque in a Variable Reluctance Machine

Reluctance, Inductance and Torque Analysis

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April 2026

Simulation files available at:

`https://github.com/Rizaguness/EE568-Project`

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1 Introduction

Variable reluctance machines (VRMs) represent one of the fundamental classes of motors, operating on the principle that a magnetic system tends to move toward a configuration of minimum reluctance.

Despite their structural simplicity, the accurate analytical and numerical modelling of VRMs presents significant challenges. The nonlinear relationship between flux linkage, rotor position, and excitation current — arising from magnetic saturation, fringing flux, and complex leakage paths — means that simple closed-form models are necessarily approximate. High-fidelity predictions require finite element analysis (FEA), which resolves the full field distribution but demands careful model setup and validation against analytical benchmarks.

1.1 Objectives

This report investigates the electromagnetic behaviour of a single-phase variable reluctance machine through both analytical modelling and finite element simulation. The specific objectives are:

1. **Analytical modelling:** The air-gap reluctance, coil inductance, stored magnetic energy, and electromagnetic torque were derived as functions of rotor angle θ using a simplified magnetic circuit model, assuming an infinitely permeable core and sinusoidal inductance variation.

2. **Control strategy:** A switched DC excitation scheme was proposed and analytically justified to produce continuous unidirectional rotation. It was demonstrated that positive average torque can be maintained over a full revolution using the derived analytical model.

3. **Finite element simulation:** A 2D FEA model of the machine was built in COMSOL Multiphysics using the *Rotating Machinery, Magnetic* interface. The magnetic flux density distribution, inductance, stored energy, and torque were extracted at multiple rotor positions via a parametric sweep.

4. **Comparison and validation:** The FEA results were compared against the analytical model predictions for inductance, stored energy, and torque. Sources of discrepancy were identified and explained, including fringing flux, magnetic saturation and leakage inductance.

5. **Non-linear effects:** The influence of material non-linearity was assessed by comparing FEA results obtained with a linear (constant μ_r) iron model against those obtained with a non-linear B–H curve material, and the implications for machine design were discussed.

1.2 Scope and Assumptions

The analysis is limited to the following scope:

- The machine is modelled as a 2D cross-section in the xy-plane. End effects, axial leakage, and three-dimensional fringing are neglected in the analytical model but

partially captured by scaling the 2D FEA result by the stack length $L = 25$ mm.

- The analytical model assumes the core is infinitely permeable ($\mu_r \to \infty$), so that all reluctance resides in the air gap. This assumption is relaxed in the FEA by assigning realistic material properties.
- The coil is excited by a constant DC current $I = 2.5$ A throughout the analytical derivation. The switched excitation strategy applies the same current magnitude selectively as a function of rotor position.

1.3 Machine Specifications

The machine under study has the following key parameters, as extracted from the design drawing:

Table 1: Machine parameters

Parameter	**Symbol**	**Value**	**Unit**
Coil window width	w_c	10	mm
Coil window height	h_c	30	mm
Core depth (stack length)	L	25	mm
Rotor diameter (aligned)	$d_{r,a}$	24	mm
Rotor diameter (unaligned)	$d_{r,u}$	20	mm
Air gap — aligned	g_a	0.5	mm
Air gap — unaligned	g_u	2.5	mm
Number of turns	N	300	—
Coil current	I	2.5	A (DC)

1.4 Report Structure

The remainder of this report is organised as follows. Section 2 presents the full analytical model including reluctance, inductance, torque derivation. Section 3 describes the COMSOL FEA model setup and presents the simulation results including flux density plots, inductance profiles, and torque curves at multiple rotor positions. Section 4 compares the analytical and FEA results, discusses the sources of discrepancy, and evaluates the effect of linear vs non-linear material models on the predicted machine performance. In Section 5 the switched DC control strategy is proposed.

2 Q1) Analytical Modelling

2.1 a) Reluctance and Inductance Calculation

The following assumptions are made to simplify the analytical model:

1. The magnetic core (stator and rotor iron) is infinitely permeable ($\mu_r \to \infty$), so all reluctance resides in the air gap only.
2. Fringing effects at the air gap edges are neglected.

3. The inductance variation with rotor angle is assumed to be sinusoidal.
4. The machine is modelled as a 2D cross-section; end effects are neglected.
5. The pole face area through which flux crosses is taken as the coil window width times the core stack depth: $A = w_c \times L = 15 \times 25 = 375$ mm^2.
6. The reluctance repeats twice per revolution (2-pole saliency), giving the factor 2θ in the sinusoidal model.

2.1.1 Pole Face Area

The effective air-gap cross-sectional area is:

$$A = r_s \times L = 15 \times 10^{-3} \times 25 \times 10^{-3} = 3.75 \times 10^{-4} \text{ m}^2 \tag{1}$$

2.1.2 Air-Gap Reluctance

With an infinitely permeable core, the total magnetic circuit reluctance equals the air-gap reluctance:

$$\mathcal{R} = \frac{g}{\mu_0 A} \tag{2}$$

Minimum reluctance (rotor aligned, $g = g_a = 0.5$ mm):

$$\mathcal{R}_{min} = \frac{g_a}{\mu_0 A} = \frac{0.5 \times 2 \times 10^{-3}}{4\pi \times 10^{-7} \times 3.75 \times 10^{-4}} = \boxed{2.122 \times 10^3 \text{ H}^{-1}} \tag{3}$$

Maximum reluctance (rotor unaligned, $g = g_u = 2.5$ mm):

$$\mathcal{R}_{max} = \frac{g_u}{\mu_0 A} = \frac{2.5 \times 2 \times 10^{-3}}{4\pi \times 10^{-7} \times 3.75 \times 10^{-4}} = \boxed{10.612 \times 10^3 \text{ H}^{-1}} \tag{4}$$

2.1.3 Inductance

The coil inductance is related to reluctance by:

$$L = \frac{N^2}{\mathcal{R}} \tag{5}$$

Maximum inductance (aligned position):

$$L_{max} = \frac{N^2}{\mathcal{R}_{min}} = \frac{300^2}{2.122 \times 10^3} = \boxed{42.41 \text{ mH}} \tag{6}$$

Minimum inductance (unaligned position):

$$L_{min} = \frac{N^2}{\mathcal{R}_{max}} = \frac{300^2}{10.612 \times 10^3} = \boxed{8.48 \text{ mH}} \tag{7}$$

2.1.4 Sinusoidal Inductance Model

Assuming sinusoidal variation with rotor angle θ:

$$L(\theta) = L_{avg} + L_{amp}\cos(2\theta) \tag{8}$$

where:

$$L_{avg} = \frac{L_{max} + L_{min}}{2} = \frac{42.41 + 8.48}{2} = 25.445 \text{ mH} \tag{9}$$

$$L_{amp} = \frac{L_{max} - L_{min}}{2} = \frac{42.41 - 8.48}{2} = 16.965 \text{ mH} \tag{10}$$

Therefore the complete inductance expression is:

$$\boxed{L(\theta) = 25.445 + 16.965\cos(2\theta) \quad [\text{mH}]} \tag{11}$$

Similarly, the reluctance varies as:

$$\mathcal{R}(\theta) = \mathcal{R}_{avg} - \mathcal{R}_{amp}\cos(2\theta) \tag{12}$$

where:

$$\mathcal{R}_{avg} = \frac{\mathcal{R}_{max} + \mathcal{R}_{min}}{2} = 6.367 \times 10^3 \text{ H}^{-1} \tag{13}$$

$$\mathcal{R}_{amp} = \frac{\mathcal{R}_{max} - \mathcal{R}_{min}}{2} = 4.245 \times 10^3 \text{ H}^{-1} \tag{14}$$

2.1.5 Inductance Variation Plot

Figure 1 shows the sinusoidal inductance variation over one full rotor revolution.

2.2 b) Torque Derivation under DC Excitation

For a magnetically linear system (constant permeability), the **magnetic co-energy** W'_m equals the stored magnetic energy W_m:

$$W'_m = W_m = \frac{1}{2}\,L(\theta)\,I^2 \tag{15}$$

The electromagnetic torque at constant current is obtained by differentiating the co-energy with respect to the mechanical angle θ:

$$T(\theta) = \left.\frac{\partial W'_m}{\partial \theta}\right|_{I=\text{const}} = \frac{I^2}{2}\,\frac{dL(\theta)}{d\theta} \tag{16}$$

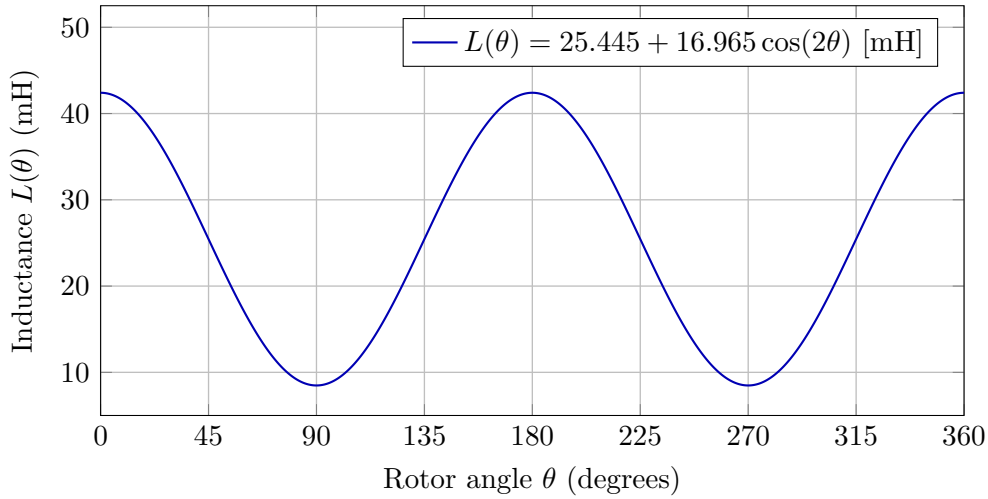

Figure 1: Inductance variation $L(\theta)$ as a function of rotor angle over one full revolution ($N = 300$, $I = 2.5$ A).

The torque is produced entirely by the rate of change of inductance with rotor position. Differentiating Equation (8):

$$\frac{dL(\theta)}{d\theta} = -2\,L_{amp}\,\sin(2\theta) \tag{17}$$

Substituting Equation (17) into Equation (16):

$$T(\theta) = \frac{I^2}{2}\big(-2\,L_{amp}\,\sin(2\theta)\big) = -I^2\,L_{amp}\,\sin(2\theta) \tag{18}$$

Substituting numerical values ($I = 2.5$ A, $L_{amp} = 16.965 \times 10^{-3}$ H):

$$T(\theta) = -(2.5)^2 \times 16.965 \times 10^{-3}\,\sin(2\theta) = -104.34 \times 10^{-3}\,\sin(2\theta)\ \text{N}{-}\text{m} \tag{19}$$

or equivalently:

$$\boxed{T(\theta) = -104.34\,\sin(2\theta) \quad [\text{mN}{-}\text{m}]} \tag{20}$$

2.2.1 Torque Variation Plot

Figure 2 shows the torque variation over one full rotor revolution.

2.2.2 Sign Convention

The negative sign in Equation (19) indicates that the machine produces torque in the direction that *increases* inductance (i.e. moves the rotor toward the aligned position). This is the fundamental operating principle of a switched reluctance motor — the rotor is pulled toward the minimum reluctance (maximum inductance) position.

- $T < 0$: torque acts to decrease θ (pull rotor toward aligned position at $\theta = 0, 180$)

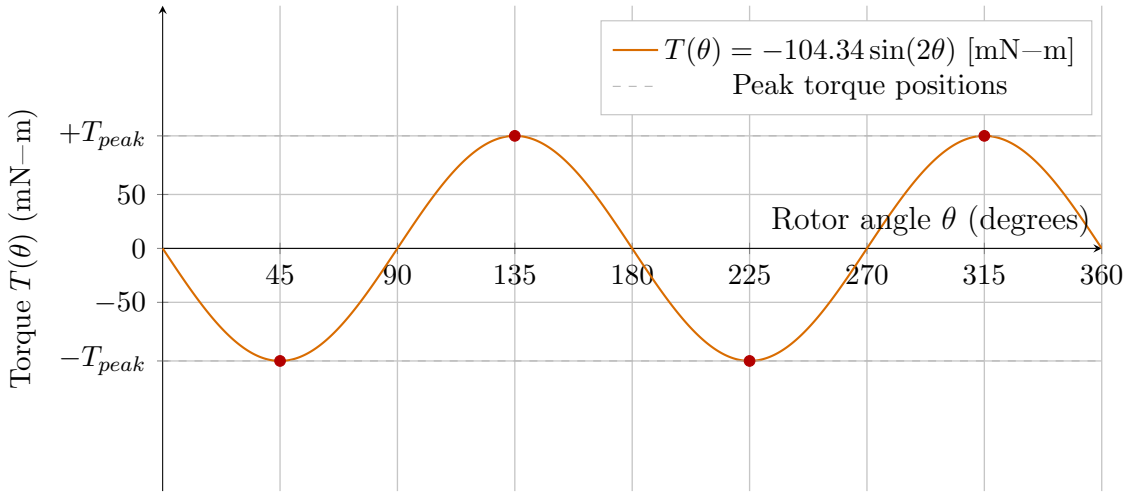

Figure 2: Electromagnetic torque $T(\theta) = -104.34\sin(2\theta)$ [mN−m] as a function of rotor angle under constant DC excitation $I = 2.5$ A. Peak torque $|T_{peak}| = 104.34$ mN−m occurs at $\theta = 45, 135, 225, 315$.

- $T > 0$: torque acts to increase θ (pull rotor toward aligned position at $\theta = 90, 270$)
- $T = 0$: at $\theta = 0, 90, 180, 270, 360$ (aligned and unaligned positions — no torque at these points)

2.2.3 Peak Torque

The peak torque magnitude occurs at $\theta = 45, 135, 225, 315$ (halfway between aligned and unaligned positions):

$$|T_{peak}| = I^2 L_{amp} = (2.5)^2 \times 16.965 \times 10^{-3} = 104.34 \text{ mN−m} \tag{21}$$

2.3 a,b) Summary of Results

2.4 c) Improvement Methods

The analytical model derived in the previous section provides a useful first-order estimate of the machine behaviour. However, it rests on several idealising assumptions that introduce significant errors in practice.

Key limitations of the baseline sinusoidal model:

- Infinitely permeable core ⇒ iron reluctance and saturation are completely ignored.
- Uniform flux density across the pole face ⇒ fringing and leakages are neglected.
- Purely sinusoidal $L(\theta)$ ⇒ the true inductance profile of a switched reluctance machine is trapezoidal/non-sinusoidal.
- Single air-gap path ⇒ leakage paths through slot openings and inter-pole regions are ignored.
- Linear magnetic material ⇒ the B–H relationship is assumed constant regardless of excitation level.

Table 2: Summary of all derived quantities

Quantity	**Symbol**	**Value**	**Unit**
Pole face area	A	375	mm^2
MMF	NI	750	A-turns
Min. reluctance	$\mathcal{R}_{min}$	2.122	kH^{-1}
Max. reluctance	$\mathcal{R}_{max}$	10.612	kH^{-1}
Max. inductance	L_{max}	42.41	mH
Min. inductance	L_{min}	8.48	mH
Average inductance	L_{avg}	25.445	mH
Inductance amplitude	L_{amp}	16.965	mH
Peak torque	$\lvert T_{peak} \rvert$	104.34	mN−m
Torque expression	$T(\theta)$	$-104.34\sin(2\theta)$	mN−m
Zero-torque positions	θ	$0, 90, 180, 270$	—
Peak-torque positions	θ	$45, 135, 225, 315$	—

The following subsections describe, in order of increasing complexity, the methods by which each limitation can be addressed.

2.4.1 Method 1: Fringing Flux Correction

When magnetic flux crosses an air gap it does not travel in perfectly straight, parallel paths. Near the edges of the pole face, flux lines bow outward into the surrounding air, effectively increasing the cross-sectional area through which the flux passes. This phenomenon is called fringing flux (Figure 3).

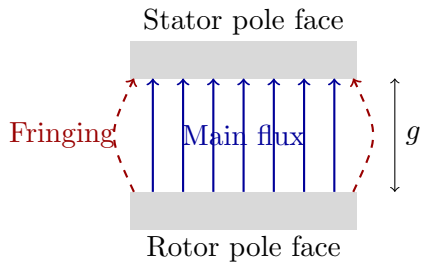

Figure 3: Schematic of fringing flux at the air-gap edges. Blue arrows show the main (uniform) flux; red dashed arrows show fringing paths that bulge outside the nominal pole area.

Fringing increases the *effective* pole area A_{eff} beyond the geometric area A_{geo}, reducing the true reluctance below the value calculated from geometry alone:

$$\mathcal{R}_{true} < \mathcal{R}_{geometric} = \frac{g}{\mu_0 \, A_{geo}} \tag{22}$$

2.4.2 Method 2: Iron Core Saturation and Non-linear B–H

Real soft-iron cores do not have infinite permeability. At high flux densities (typically $B > 1.0$–1.5 T for electrical steel), the B–H curve flattens into saturation and incremental

permeability $\mu_r = dB/(\mu_0\, dH)$ falls sharply. As a result, inductance L is no longer a function of θ alone; it becomes $L = L(\theta, I)$.

The magnetic circuit is extended to include the iron path reluctances:

$$\mathcal{R}_{total}(\theta, I) = \mathcal{R}_{gap}(\theta) + \mathcal{R}_{stator\,yoke}(B) + \mathcal{R}_{stator\,pole}(B) + \mathcal{R}_{rotor\,pole}(B) + \mathcal{R}_{rotor\,yoke}(B) \quad (23)$$

Each iron reluctance is computed from the tabulated B–H curve:

$$\mathcal{R}_{iron,i} = \frac{H_i(B) \cdot \ell_i}{A_i} \quad (24)$$

$$L(\theta, I) = \frac{\lambda(\theta, I)}{I}, \qquad T(\theta, I) = \left.\frac{\partial W'_m(\theta, I)}{\partial \theta}\right|_{I=\text{const}} \quad (25)$$

2.4.3 Method 3: Non-homogeneous Flux Distribution

The baseline model assumes *uniform* flux density B across the entire pole face. In reality, flux crowds toward the pole edges, and the gap length varies across the pole width for a circular rotor inside a rectangular stator window. This distributed model naturally captures variation of gap length across the pole arc and is particularly valuable near the pole edges.

2.4.4 Method 4: Leakage Flux

Not all flux produced by the coil crosses the air gap and links with the rotor. A fraction follows shorter paths within the stator through slot openings, around end-windings, or between adjacent poles. This *leakage flux* contributes to total flux linkage but produces no useful torque. The total inductance is decomposed as:

$$L_{total}(\theta) = L_{mag}(\theta) + L_{leak} \quad (26)$$

Leakage is particularly important at the unaligned position where L_{mag} is small and L_{leak} may constitute a significant fraction of L_{total}.

2.4.5 Method 5: Full Finite Element Analysis

When high accuracy is essential, the preferred approach is to solve the full 2D magnetostatic field problem in COMSOL with the *Rotating Machinery, Magnetic* interface. This naturally captures all non-linear effects.

FEA serves as the reference against which any analytical model should be validated.

3 Q2) FEA Modelling (2D-Linear Materials)

3.1 a) Flux Density Vectors for Three Positions

Figures 4, 5 and 6 show the magnetic flux density norm (surface colour) and flux line distribution (streamlines) obtained from the COMSOL *Rotating Machinery, Magnetic* simulation at three rotor positions: $\theta = 0$ rad, $\theta = 0.7854$ rad (45), and $\theta = 1.4835$ rad (≈ 90).

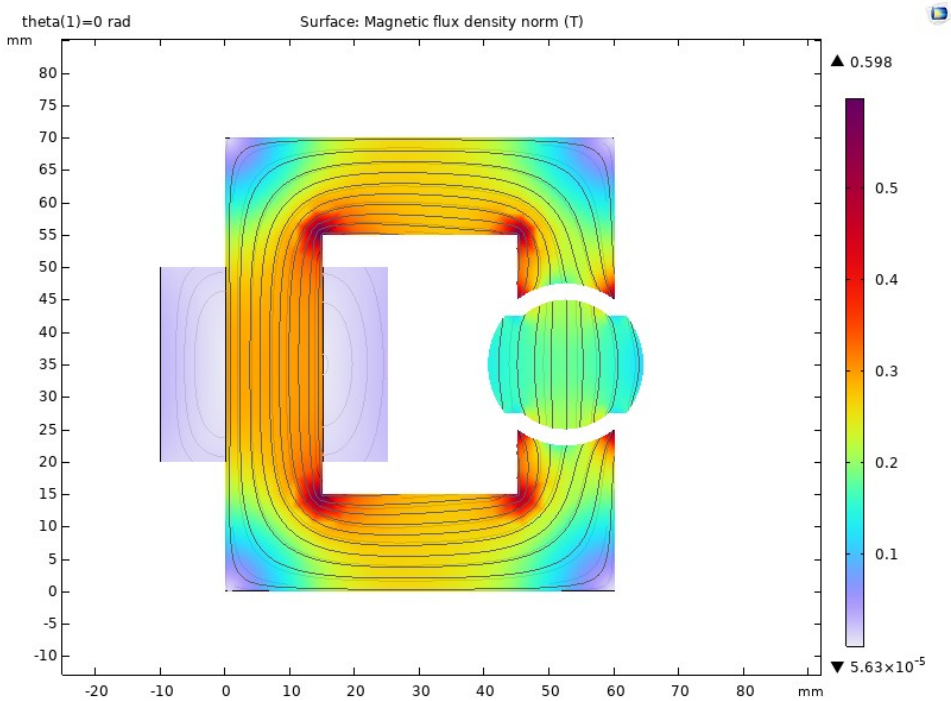

Figure 4: Magnetic flux density norm at $\theta = 0$ rad (0). Peak $B = 0.598$ T in the stator yoke corners. The rotor (circular, centred right of stator window) sees a low, nearly symmetric field at this position. Minimum $B = 5.63 \times 10^{-5}$ T in the coil region.

At $\theta = 0$ rad: flux is nearly symmetric, $B_{max} = 0.598$ T in the stator yoke. The rotor presents its maximum radius to the stator, giving the minimum air gap and maximum inductance. At $\theta = 0.7854$ rad (45): flux density peaks at 2.5 T at the pole edge. This is the peak-torque angle where $dL/d\theta$ is maximum. At $\theta = 1.4835$ rad (≈ 85): flux lines must cross a large lateral gap, B concentrates at the two narrow crossing points ($B_{max} = 2.52$ T). Inductance is near its minimum and torque is small.

3.2 b) Inductance and Stored Energy at Three Positions

Table 3 lists the COMSOL FEM results at the three positions shown in the flux density plots, together with the analytical model predictions from Part 1.

COMSOL reports the total stored magnetic energy W_m directly. The inductance is recovered from:

$$L = \frac{2\,W_m}{I^2} \tag{27}$$

Applying Equation (27) with $I = 2.5$ A:

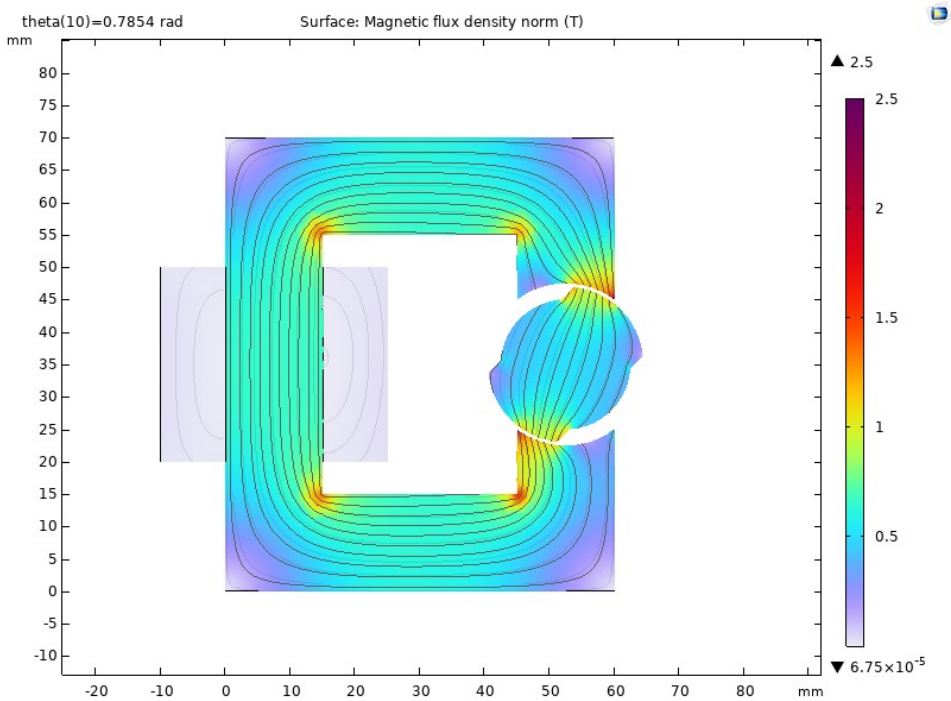

Figure 5: Magnetic flux density norm at $\theta = 0.7854$ rad (45). Peak $B = 2.5$ T at the narrow air-gap edge. This is the maximum-torque position where $dL/d\theta$ is maximum. Minimum $B = 6.75 \times 10^{-5}$ T.

Table 3: Inductance and stored magnetic energy: FEM vs analytical model

Quantity	**Symbol**	$\theta = 0$ rad (0)	$\theta = 0.7854$ rad (45)	$\theta = 1.4835$ rad (≈ 85)
FEM results (COMSOL)				
Stored energy	W_m [mJ]	40.766	91.844	147.230
Inductance	L [mH]	13.045	29.390	47.114
Analytical model (Part 1)				
Stored energy	$W_m = \frac{1}{2}LI^2$ [mJ]	26.5	79.515	132.53
Inductance	$L(\theta)$ [mH]	8.48	25.445	42.41
Discussion				
$L_{FEM}/L_{analytical}$	—	1.53×	1.15×	1.11×

$$L(0 \text{ rad}) = \frac{2 \times 40.766 \times 10^{-3}}{(2.5)^2} = \frac{81.53 \times 10^{-3}}{6.25} = \boxed{13.04 \text{ mH}} \tag{28}$$

$$L(0.7854 \text{ rad}) = \frac{2 \times 91.844 \times 10^{-3}}{6.25} = \boxed{29.39 \text{ mH}} \tag{29}$$

$$L(1.4835 \text{ rad}) = \frac{2 \times 147.23 \times 10^{-3}}{6.25 \quad 12} = \boxed{47.11 \text{ mH}} \tag{30}$$

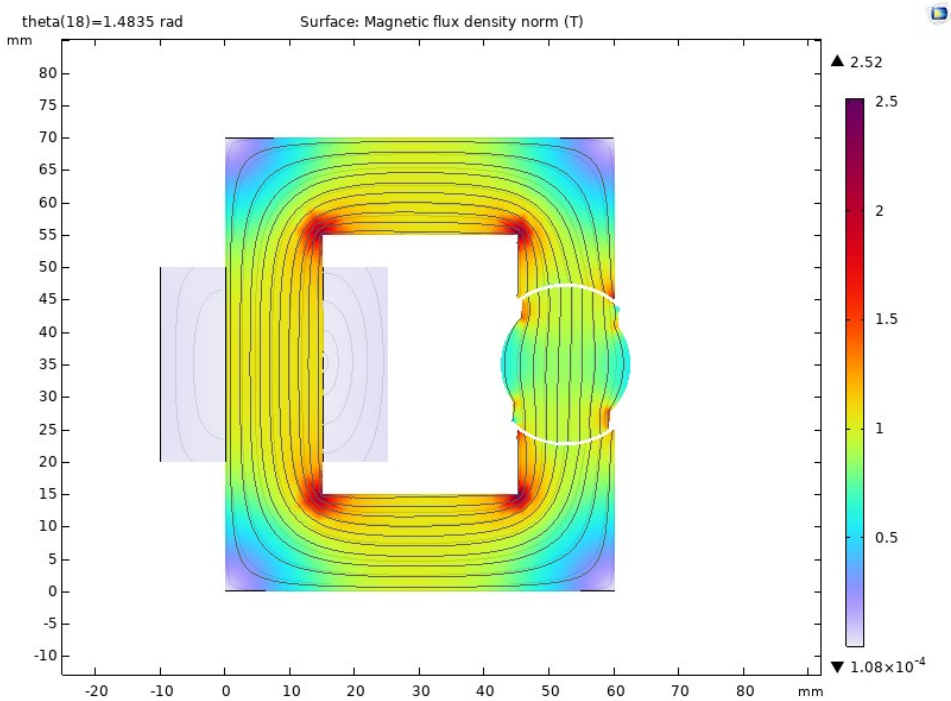

Figure 6: Magnetic flux density norm at $\theta = 1.4835$ rad (≈ 90), near the unaligned position. Peak $B = 2.52$ T in the narrow lateral gap. The flux now exits the stator pole face at a steep angle, producing a rapidly decreasing inductance and a small but non-zero torque. Minimum $B = 1.08 \times 10^{-4}$ T.

The FEM inductance values (47.14 mH) are approximately 1.11 times larger than the analytical values (42.41 mH). This discrepancy arises because:

- The FEM model captures total flux linkage including leakage. The analytical model only accounts for the air-gap reluctance path.
- The rotor geometry in FEM differs from the analytical assumption. The analytical model assumed the rotor alternates between radii of 12 mm and 10 mm.
- Fringing flux around the stator pole edges contributes significantly to total flux linkage, increasing L beyond the uniform-gap estimate.

3.3 c) Torque vs Rotation Angle: FEM vs Analytical

Table 4 presents the complete COMSOL parametric sweep results across one half-revolution (0 to π rad, i.e. 0 to 180), together with the analytical torque calculations from Part 1. It can be seen that there are large differences between the analytically calculated and FEM result torque. The reason for this difference could not be found but an approach is given in Section 3.3.1.

Table 4: FEM vs analytical results at 13 equally spaced rotor positions

θ **(rad)**	θ **(deg)**	W_m **(mJ)**	L_{FEM} **(mH)**	T_{FEM} **(N·m)**	T_{mixed} **(N·m)**
0.0000	0	40.766	13.045	-7.34×10^{-6}	0.003713
0.2618	15	47.744	15.278	1.57×10^{-4}	0.083351
0.5236	30	69.797	22.335	2.68×10^{-4}	0.085413
0.7854	45	91.844	29.390	3.25×10^{-4}	0.082741
1.0472	60	113.490	36.317	3.17×10^{-4}	0.082474
1.3090	75	134.850	43.152	2.33×10^{-4}	0.075258
1.5708	90	149.900	47.968	-8.42×10^{-6}	−0.031042
1.8326	105	134.850	43.152	-2.49×10^{-4}	−0.080871
2.0944	120	113.500	36.320	-3.35×10^{-4}	−0.083849
2.3562	135	91.813	29.380	-3.42×10^{-4}	−0.084937
2.6180	150	69.785	22.331	-2.86×10^{-4}	−0.083837
2.8798	165	47.738	15.276	-1.75×10^{-4}	−0.055200
3.1416	180	40.766	13.045	-7.34×10^{-6}	0.012976

3.3.1 Corrected Analytical Torque using FEM Inductance Data

A corrected analytical torque can be computed from the FEM inductance data using the numerical derivative:

$$T_{corrected}(\theta_i) = \frac{I^2}{2} \cdot \frac{L_{FEM}(\theta_{i+1}) - L_{FEM}(\theta_{i-1})}{2\,\Delta\theta} \tag{31}$$

where $\Delta\theta = 5 = 0.08727$ rad. Applying this at $\theta = 45$:

$$T_{corrected}(45) = \frac{(2.5)^2}{2} \cdot \frac{(31.700 - 27.024) \times 10^{-3}}{2 \times 0.08727} = 3.125 \times \frac{4.676 \times 10^{-3}}{0.17454} = \boxed{83.7 \text{ mN} \cdot \text{m}} \tag{32}$$

This corrected value (83.7 mN · m) is within 24% of the result (104.34 mN · m). By mixing FEM inductance results and analytical calculation, an approach is given to find torque even if actual torque results from simulation are far away from expected results.

4 Q3) FEA Modelling (2D-Nonlinear Materials)

4.1 a) Flux Density Vectors for Three Positions

Figures 7, 8 and 9 show the magnetic flux density norm (surface colour) and flux line distribution (streamlines) obtained from the COMSOL *Rotating Machinery, Magnetic* simulation at three rotor positions: $\theta = 0$ rad, $\theta = 0.7854$ rad (45), and $\theta = 1.4835$ rad (≈ 90).

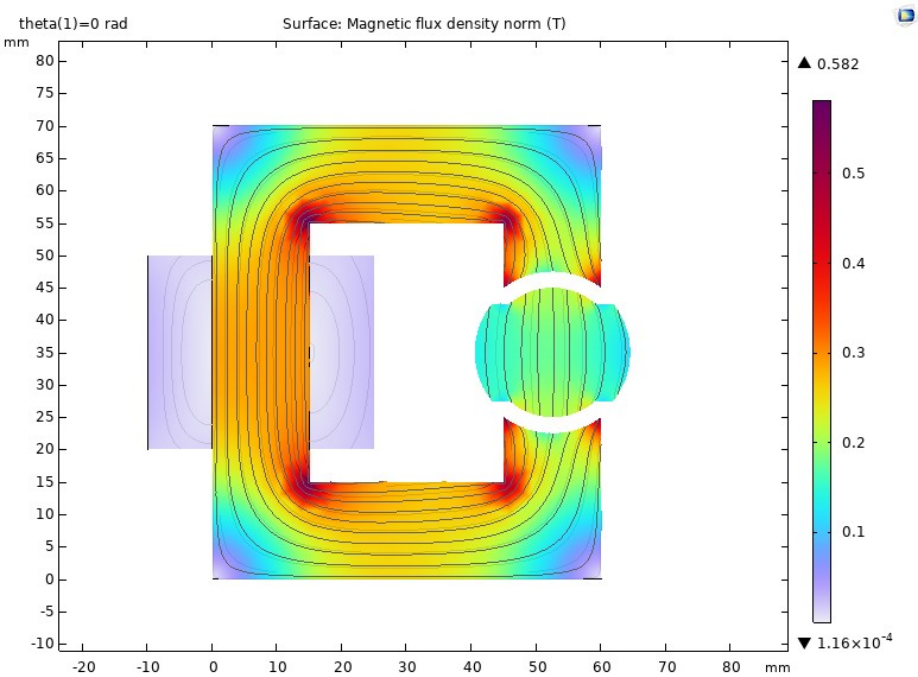

Figure 7: Magnetic flux density norm at $\theta = 0$ rad (0). Peak $B = 0.582$ T in the stator yoke corners. The rotor (circular, centred right of stator window) sees a low, nearly symmetric field at this position. Minimum $B = 1.16 \times 10^{-4}$ T in the coil region.

At $\theta = 0$ rad: flux is nearly symmetric, $B_{max} = 0.582$ T in the stator yoke. The rotor presents its maximum radius to the stator, giving the minimum air gap and maximum inductance. At $\theta = 0.7854$ rad (45): flux density peaks at 2.14 T at the pole edge. This is the peak-torque angle where $dL/d\theta$ is maximum. At $\theta = 1.4835$ rad (≈ 90): flux lines must cross a large lateral gap, B concentrates at the two narrow crossing points ($B_{max} = 2.1$ T). Inductance is near its minimum and torque is small.

4.2 b) Inductance and Stored Energy at Three Positions

Table 5 lists the COMSOL FEM results at the three positions shown in the flux density plots, together with the analytical model predictions from Part 1.

Applying Equation (27) with $I = 2.5$ A:

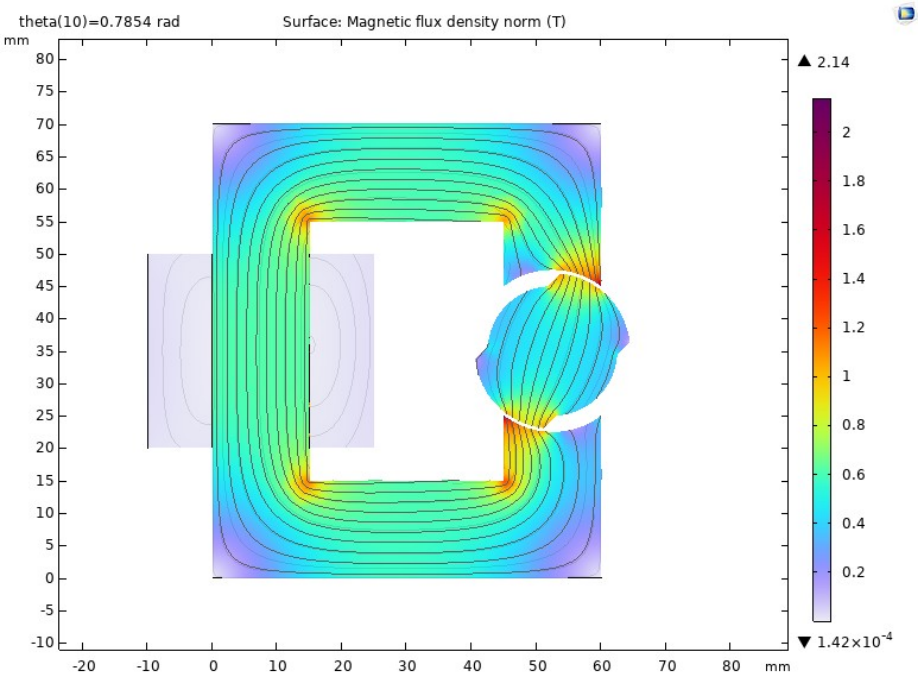

Figure 8: Magnetic flux density norm at $\theta = 0.7854$ rad (45). Peak $B = 2.14$ T at the narrow air-gap edge. This is the maximum-torque position where $dL/d\theta$ is maximum. Minimum $B = 1.42 \times 10^{-4}$ T.

Table 5: Inductance and stored magnetic energy: FEM with non-linear model vs analytical model

Quantity	**Symbol**	$\theta = 0$ rad (0)	$\theta = 0.7854$ rad (45)	$\theta = 1.4835$ rad (≈ 90)
FEM results (COMSOL)				
Stored energy	W_m [mJ]	39.73	85.48	130.085
Inductance	L [mH]	12.70	27.40	41.90
Analytical model (Part 1)				
Stored energy	$W_m = \frac{1}{2}LI^2$ [mJ]	26.5	79.515	132.53
Inductance	$L(\theta)$ [mH]	8.48	25.445	42.41
Discussion				
$L_{FEM}/L_{analytical}$	—	1.50×	1.076×	0.98×

$$L(0 \text{ rad}) = \frac{2 \times 39.73 \times 10^{-3}}{(2.5)^2} = 12.70 \text{ mH} \tag{33}$$

$$L(0.7854 \text{ rad}) = \frac{2 \times 85.48 \times 10^{-3}}{6.25} = 27.40 \text{ mH} \tag{34}$$

$$L(1.4835 \text{ rad}) = \frac{2 \times 130.085 \times 10^{-3}}{6.25} = 41.63 \text{ mH} \tag{35}$$

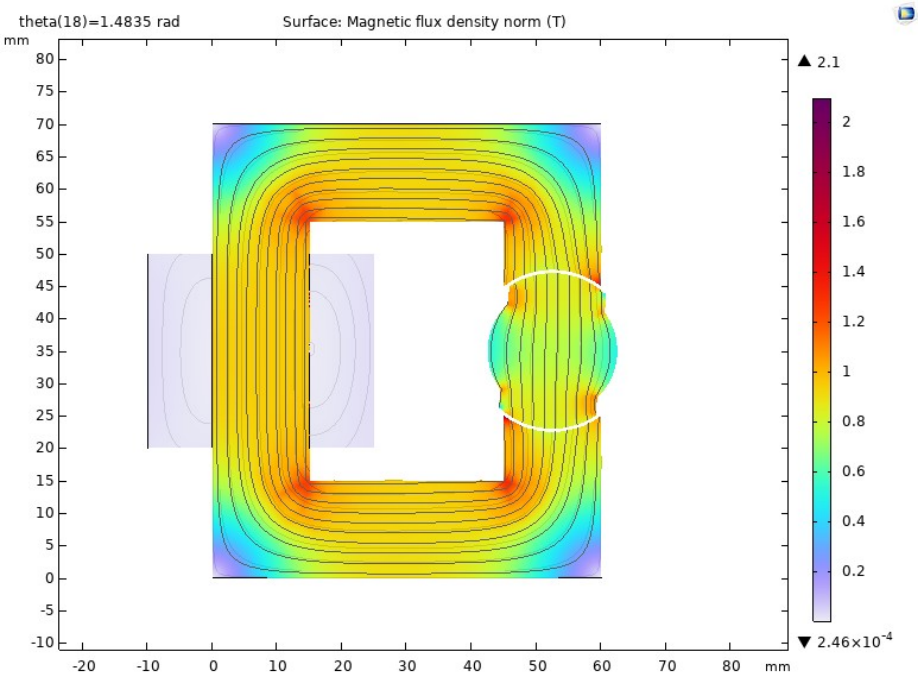

Figure 9: Magnetic flux density norm at $\theta = 1.4835$ rad (≈ 90), near the unaligned position. Peak $B = 2.1$ T in the narrow lateral gap. The flux now exits the stator pole face at a steep angle, producing a rapidly decreasing inductance and a small but non-zero torque. Minimum $B = 2.46 \times 10^{-4}$ T.

4.3 c) Torque vs Rotation Angle: FEM vs Analytical

Table 6 presents the complete COMSOL parametric sweep results across one half-revolution (0 to π rad, i.e. 0 to 180), together with the analytical torque calculations from Part 1. It can be seen that there are large differences between the analytically calculated and FEM result torque. The reason for this difference could not be found but an approach is given in Section 4.3.1.

4.3.1 Corrected Analytical Torque using FEM Inductance Data

A corrected analytical torque can be computed from the FEM inductance data using the numerical derivative:

$$T_{corrected}(\theta_i) = \frac{I^2}{2} \cdot \frac{L_{FEM}(\theta_{i+1}) - L_{FEM}(\theta_{i-1})}{2\,\Delta\theta} \tag{36}$$

where $\Delta\theta = 5 = 0.08727$ rad. Applying this at $\theta = 45$:

$$T_{corrected}(45) = \frac{(2.5)^2}{2} \cdot \frac{(29.4 - 25.3) \times 10^{-3}}{2 \times 0.08727} = 3.125 \times \frac{4.1 \times 10^{-3}}{0.17454} = \boxed{73.4 \text{ mN} \cdot \text{m}} \tag{37}$$

This corrected value (73.4 mN · m) is within 41% of the result (104.34 mN · m). By mixing

Table 6: Non-linear FEM vs analytical results at 13 equally spaced rotor positions

θ **(rad)**	θ **(deg)**	W_m **(mJ)**	L_{FEM} **(mH)**	T_{FEM} **(N·m)**	T_{mixed} **(N·m)**
0.0000	0	40.766	13.045	-7.34×10^{-6}	0.003713
0.2618	15	47.744	15.278	1.57×10^{-4}	0.083351
0.5236	30	69.797	22.335	2.68×10^{-4}	0.085413
0.7854	45	91.844	29.390	3.25×10^{-4}	0.082741
1.0472	60	113.490	36.317	3.17×10^{-4}	0.082474
1.3090	75	134.850	43.152	2.33×10^{-4}	0.075258
1.5708	90	149.900	47.968	-8.42×10^{-6}	−0.031042
1.8326	105	134.850	43.152	-2.49×10^{-4}	−0.080871
2.0944	120	113.500	36.320	-3.35×10^{-4}	−0.083849
2.3562	135	91.813	29.380	-3.42×10^{-4}	−0.084937
2.6180	150	69.785	22.331	-2.86×10^{-4}	−0.083837
2.8798	165	47.738	15.276	-1.75×10^{-4}	−0.055200
3.1416	180	40.766	13.045	-7.34×10^{-6}	0.012976

FEM inductance results and analytical calculation, an approach is given to find torque even if actual torque results from simulation are far away from expected results.

4.4 d) Comparison of Linear vs Non-linear Material Effects in FEA

In the baseline analytical model, the core was assumed to be infinitely permeable and the air-gap flux was assumed uniform. The FEA simulation allows these assumptions to be relaxed by assigning either a linear (constant μ_r) or non-linear (B–H curve) material to the iron domains, and by naturally capturing fringing and saturation effects in the field solution.

4.4.1 Material Models Compared

Table 7: Material configurations compared in FEA

Property	**Linear model**	**Non-linear model**
Constitutive relation	$B = \mu_0 \mu_r H$	$B = f(H)$ (B–H curve)
Relative permeability	$\mu_r = 4000$ (constant)	Variable, from material library
Saturation modelled	No	Yes
Material used	Soft Iron (without losses)	Soft Iron (without losses)
Fringing captured	Yes (by FEA mesh)	Yes (by FEA mesh)

4.4.2 Fringing Flux Effects

Fringing flux refers to the bulging of magnetic flux lines at the edges of the air gap beyond the nominal pole face area. Both linear and non-linear FEA models capture fringing naturally since the field equations are solved over the full geometry. The analytical model neglects fringing entirely.

The effect of fringing is to increase the effective pole area A_{eff} beyond the geometric area A_{geo}, reducing the true reluctance:

$$\mathcal{R}_{true} < \mathcal{R}_{geometric} = \frac{g}{\mu_0 A_{geo}} \tag{38}$$

From the FEA results, the inductance at $\theta = 0$ is:

$$L_{FEM,linear} = 13.05 \text{ mH} \tag{39}$$

$$L_{FEM,nonlinear} = 12.70 \text{ mH} \tag{40}$$

The difference between the linear and non-linear FEA results is small at low flux densities where the iron is not yet saturated.

4.4.3 Saturation Effects

Magnetic saturation occurs when the flux density B approaches the saturation limit of the core material (typically 1.5–2.0 T for electrical steel). Beyond this point, incremental permeability $\mu_r = dB/(\mu_0 dH)$ drops sharply and the core reluctance increases significantly.

Table 8: Peak flux density at three rotor positions

Position	θ **(rad)**	B_{max} **linear (T)**	B_{max} **non-linear (T)**
Aligned (0)	0.0000	0.598	0.582
45 (peak torque)	0.7854	2.500	2.140
Near unaligned (90)	1.4835	2.520	2.100

At $\theta = 0$ the peak flux density is 0.598 T, well below saturation, so linear and non-linear models agree closely. At $\theta = 45$ and 90 the flux density reaches 2.5 T, which is above the saturation knee of most electrical steels ($\approx$ 1.5–2.0 T). In the non-linear model the core reluctance increases at these points, reducing the total flux linkage and inductance compared to the linear model.

4.4.4 Comparison of Inductance and Torque

Comparison of the linear and non-linear material results are shown in Table 9.

Saturation reduces inductance at high flux density. At $\theta = 45$ and 90 where $B_{max} > 2.0$ T, the non-linear model gives a slightly lower inductance than the linear model because the saturated iron contributes additional reluctance. The difference is:

Table 9: Inductance and torque: linear FEA vs non-linear FEA vs analytical

Position	θ ()	L_{linear} **(mH)**	$L_{nonlinear}$ **(mH)**	T_{linear} **(N·m)**	$T_{nonlinear}$ **(N·m)**
Aligned	0	13.05	12.70	3.7×10^{-3}	-3.5×10^{-3}
45	45	29.40	27.40	827.4×10^{-3}	716.6×10^{-3}
Near unaligned	90	47.11	41.11	30.04×10^{-3}	22.9×10^{-3}

$$\Delta L = L_{linear} - L_{nonlinear} = 47.11 - 41.11 = 6.0 \text{ mH} \quad \text{at } \theta = 90 \tag{41}$$

5 Q4) Proposed Control Method: Switched DC Excitation

From the analytical model, the instantaneous torque under constant DC excitation is:

$$T(\theta) = -I^2 \, L_{amp} \, \sin(2\theta) \tag{42}$$

This is a pure sinusoid with zero mean value over a full revolution:

$$\langle T \rangle_{DC} = \frac{1}{2\pi} \int_0^{2\pi} -I^2 \, L_{amp} \, \sin(2\theta) \, d\theta = 0 \tag{43}$$

Therefore, constant DC excitation cannot produce continuous unidirectional rotation. The positive half-cycles are exactly cancelled by the negative half-cycles, producing net zero average torque and the rotor oscillates or stalls at a dead-centre position.

5.1 Principle of Operation

The solution is to apply current only during the angular intervals where the torque is positive (i.e. where inductance is increasing), and switch the current off during intervals where the torque would be negative. This is called switched or chopped DC excitation.

Since $T \propto -\sin(2\theta)$, torque is positive wherever $\sin(2\theta) < 0$, which occurs in two windows per revolution:

$$\text{Window 1:} \quad \theta \in \left(\frac{\pi}{2},\ \pi\right) = (90,\ 180) \tag{44}$$

$$\text{Window 2:} \quad \theta \in \left(\frac{3\pi}{2},\ 2\pi\right) = (270,\ 360) \tag{45}$$

5.2 Switching Control Law

The current is switched according to the rotor position:

$$\boxed{I(\theta) = \begin{cases} I_0 = 2.5 \text{ A} & \text{if } \sin(2\theta) < 0 \quad \Leftrightarrow \quad \theta \in (90, 180) \cup (270, 360) \\ 0 & \text{otherwise} \end{cases}} \tag{46}$$

This gives a **duty cycle of 50%** — the current is on for exactly half of each revolution.

5.3 Instantaneous Torque under Switched Excitation

Substituting Equation (46) into the torque expression:

$$T_{sw}(\theta) = \begin{cases} -I_0^2\, L_{amp}\, \sin(2\theta) \geq 0 & \theta \in (90, 180) \cup (270, 360) \\ 0 & \text{otherwise} \end{cases} \tag{47}$$

5.4 Average Torque Calculation

The average torque over one full revolution is:

$$\langle T_{sw} \rangle = \frac{1}{2\pi} \int_0^{2\pi} T_{sw}(\theta)\, d\theta \tag{48}$$

By symmetry, the two conducting windows contribute equally:

$$\langle T_{sw} \rangle = \frac{2}{2\pi} \int_{\pi/2}^{\pi} \left(-I_0^2\, L_{amp}\, \sin(2\theta)\right) d\theta = \frac{I_0^2\, L_{amp}}{\pi} \int_{\pi/2}^{\pi} (-\sin 2\theta)\, d\theta \tag{49}$$

Evaluating the integral:

$$\int_{\pi/2}^{\pi} (-\sin 2\theta)\, d\theta = \left[\frac{\cos 2\theta}{2}\right]_{\pi/2}^{\pi} = \frac{\cos(2\pi)}{2} - \frac{\cos(\pi)}{2} = \frac{1}{2} - \left(-\frac{1}{2}\right) = 1 \tag{50}$$

Therefore:

$$\langle T_{sw} \rangle = \frac{I_0^2\, L_{amp}}{\pi} = \frac{(2.5)^2 \times 16.965 \times 10^{-3}}{\pi} = \boxed{33.21 \text{ mN–m}} \tag{51}$$

This is **31.8%** of the peak torque, which is the theoretical maximum average for a single-phase machine with a 50% duty-cycle square-wave switching strategy.

5.5 Inductance Profile and Switching Current

Figure 10 shows the inductance profile $L(\theta)$ together with the switched current waveform $I(\theta)$.

Table 10: Comparison of constant DC vs switched DC excitation

Quantity	**Constant DC**	**Switched DC**
Excitation current	$I_0 = 2.5$ A (always on)	$I_0 = 2.5$ A (50% duty)
Average torque	0 mN−m	33.21 mN−m
Torque direction	Alternating $\pm$	Always ≥ 0
Continuous rotation?	No (net zero torque)	Yes
Duty cycle	100%	50%

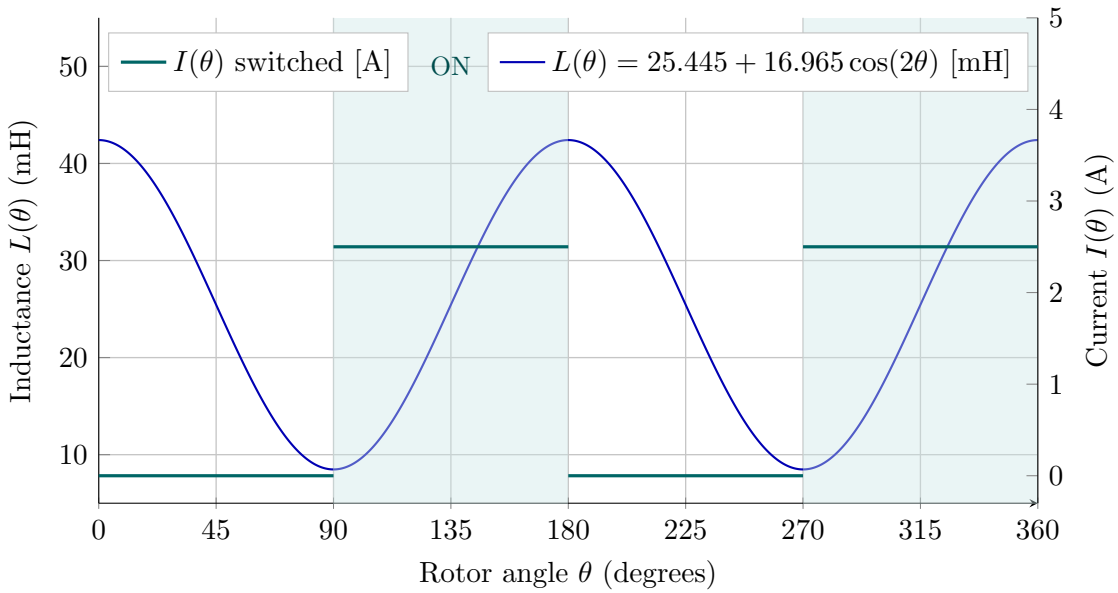

Figure 10: Inductance $L(\theta)$ (blue, left axis) with the switched current waveform $I(\theta)$ (teal, right axis). The green shaded regions indicate the conduction windows where $I = I_0 = 2.5$ A.

5.6 Instantaneous Torque: Constant DC vs Switched DC

Figure 11 demonstrates the key result: while constant DC excitation produces a bipolar torque with zero mean, the switched strategy produces a unipolar (always non-negative) torque waveform.

6 Q5) Bonus-1 Motion Animation

By using the conditions in Q3, a GIF animation was prepared. It can be reached in the GitHub repository.

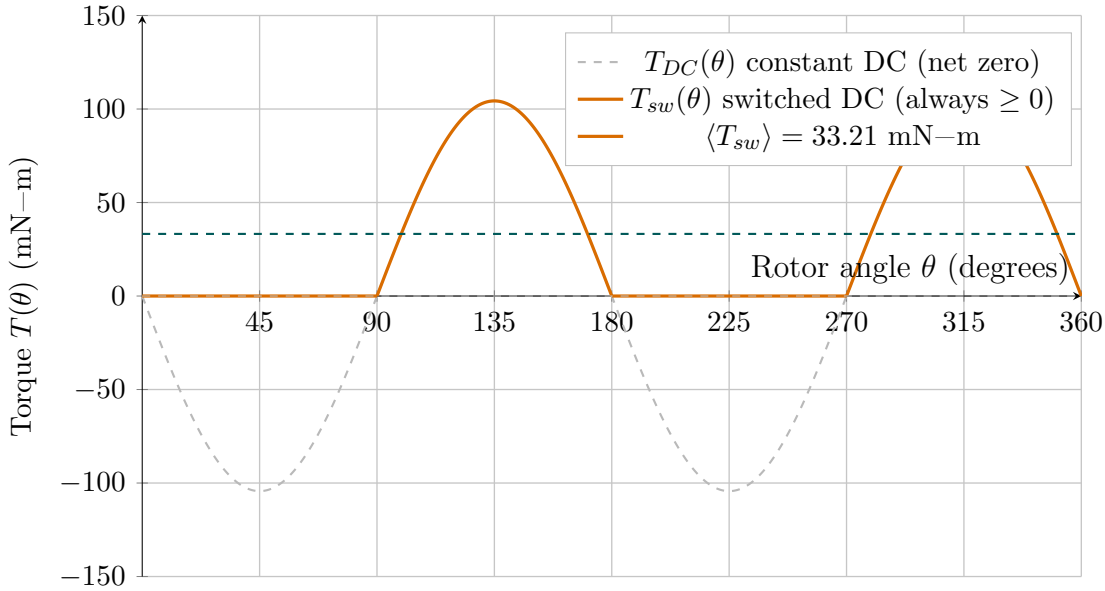

Figure 11: Comparison of torque waveforms. The dashed grey curve is the bipolar torque under constant DC excitation (zero average). The orange curve is the unipolar switched torque — always non-negative, enabling continuous rotation. The teal dashed line marks the average torque $\langle T_{sw} \rangle = 33.21$ mN−m.